

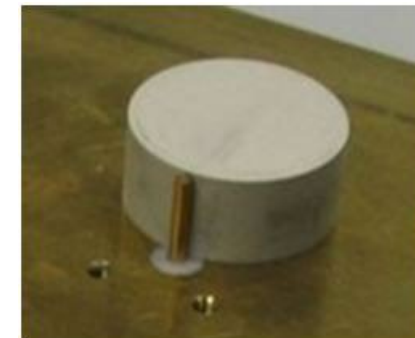
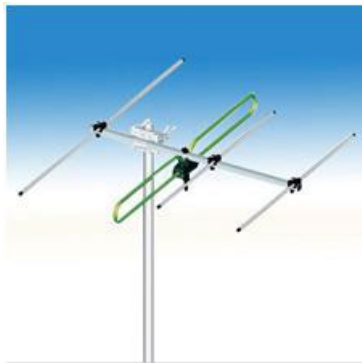
ECE 3317

Applied Electromagnetic Waves

Prof. David R. Jackson
Fall 2025

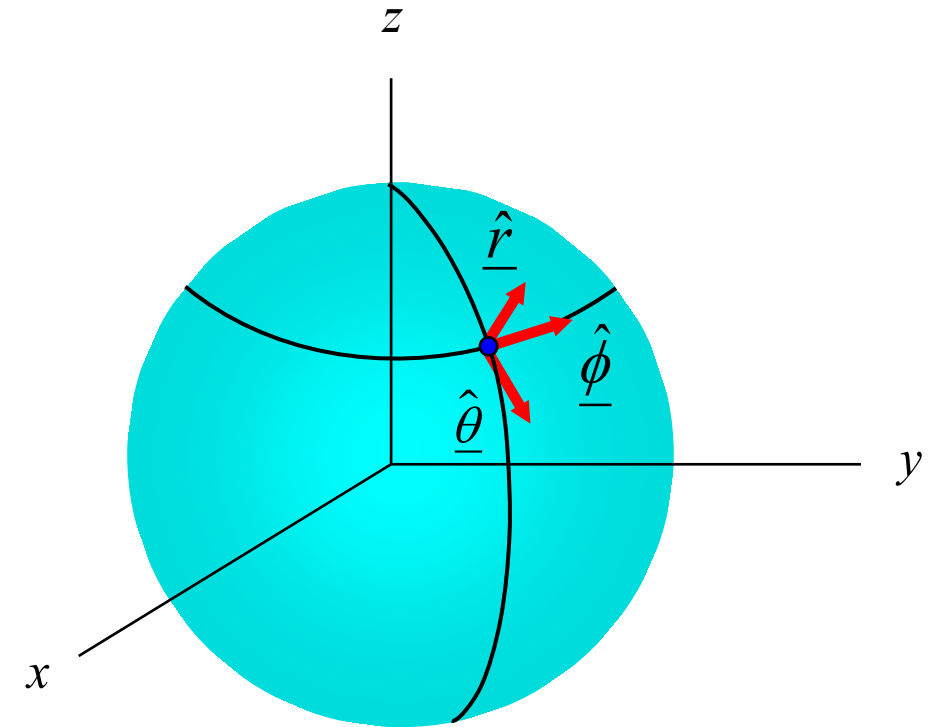
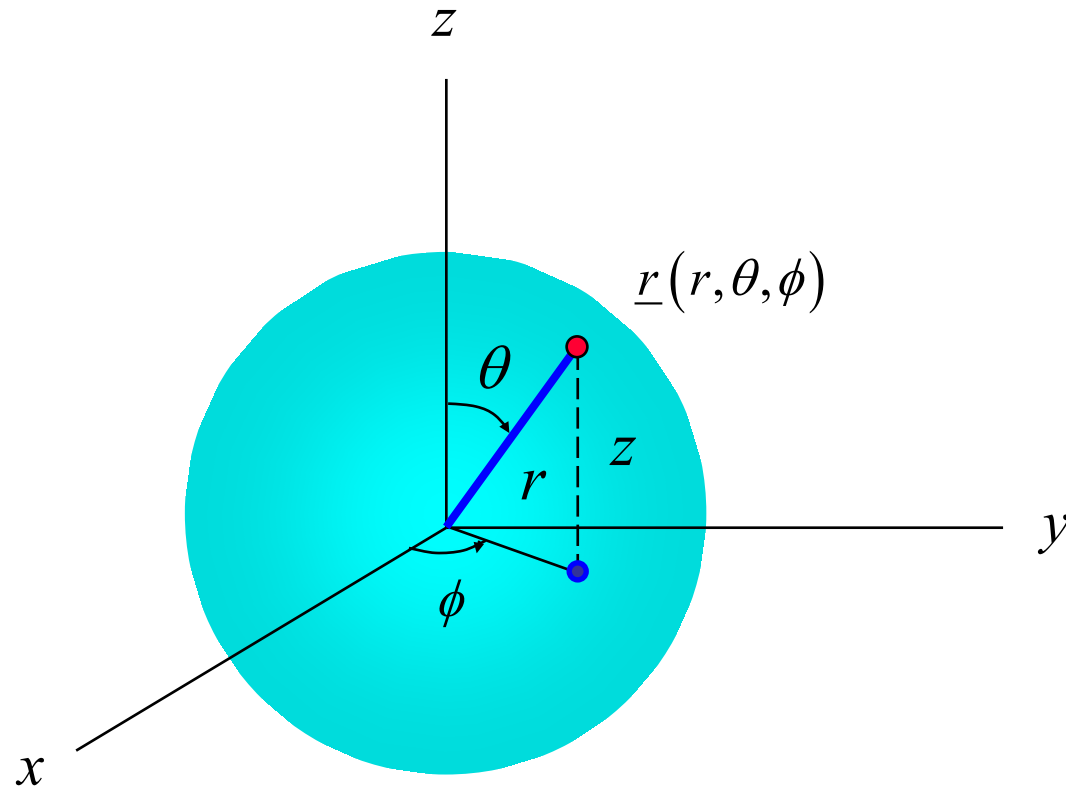
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Antenna Properties



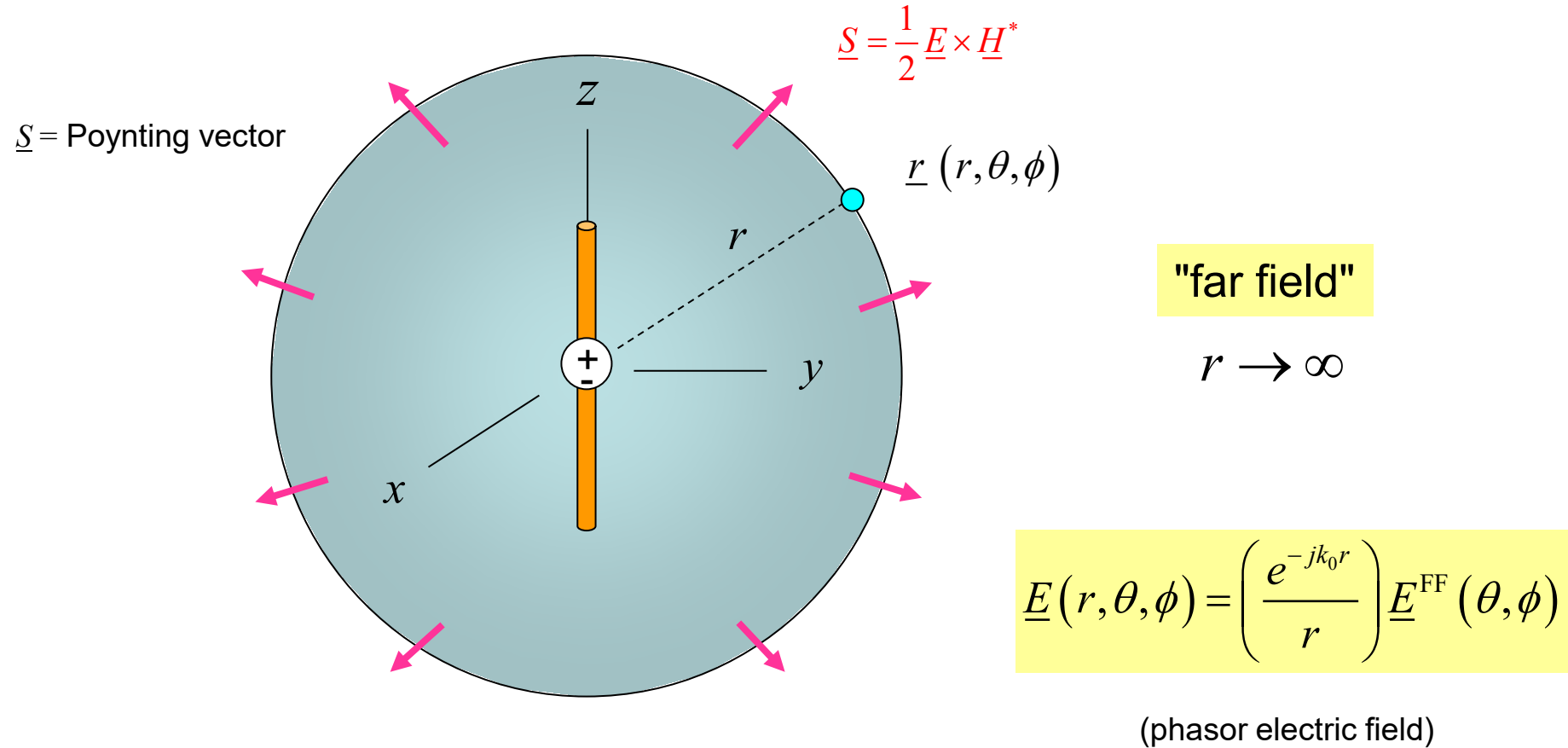
Spherical Coordinates

Radiation patterns of an antenna are usually described in spherical coordinates.



Antenna Radiation

We consider here the radiation from an antenna.

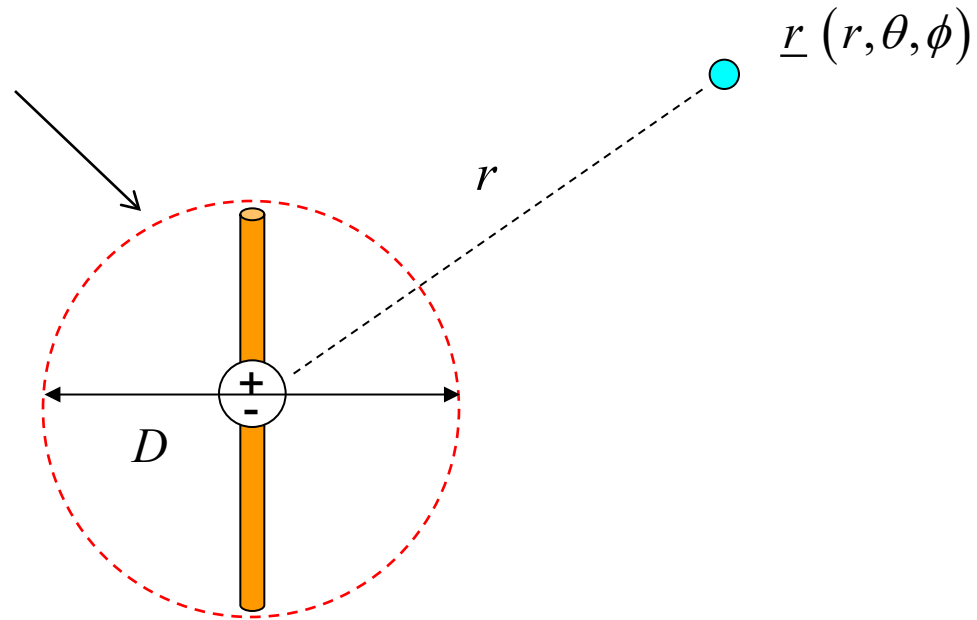


- The far-field radiation acts like a plane wave going in the radial direction.
- The shape of the pattern in the far field is only a function of (θ, ϕ) .

Antenna Radiation (cont.)

How far do we have to go to be in the far field?

Sphere of minimum diameter D that encloses the antenna.

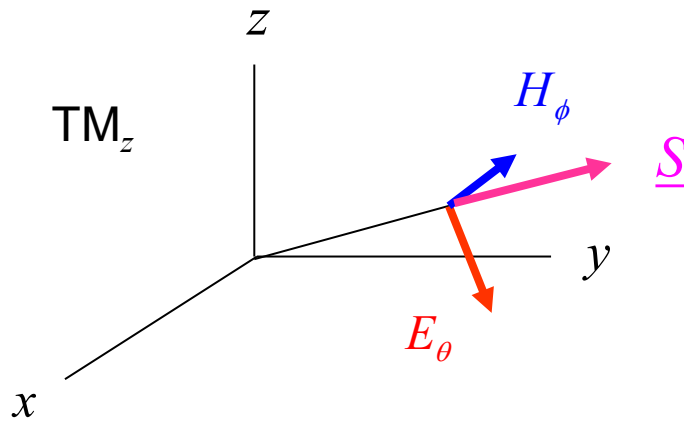


$$r > \frac{2D^2}{\lambda_0}$$

Antenna Radiation (cont.)

The far-field has the following form:

Example: vertical dipole

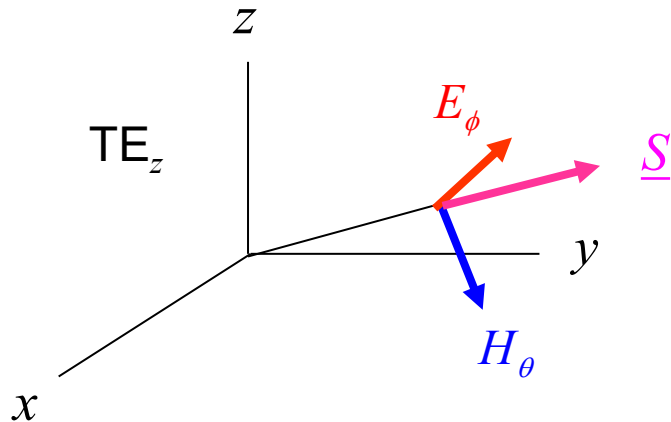


$$\frac{E_{\theta}}{H_{\phi}} = \eta_0$$

$$\underline{E} = \hat{\theta} E_{\theta} + \hat{\phi} E_{\phi} \text{ (phasor electric field vector)}$$

$$\underline{H} = \hat{\theta} H_{\theta} + \hat{\phi} H_{\phi} \text{ (phasor magnetic field vector)}$$

Example: horizontal loop antenna



$$\frac{E_{\phi}}{H_{\theta}} = -\eta_0$$

Depending on the type of antenna, either or both polarizations may be radiated (e.g., a vertical wire antenna radiates only E_{θ} polarization).

Antenna Radiation (cont.)

The power density in the far field is:

$$\underline{S}(r, \theta, \phi) = \underline{\hat{r}} \left(|E_\theta|^2 + |E_\phi|^2 \right) \left(\frac{1}{2\eta_0} \right)$$

or

$$\underline{S}(r, \theta, \phi) = \underline{\hat{r}} \left(\frac{|\underline{E}|^2}{2\eta_0} \right)$$

Antenna Radiation (cont.)

The far field always has the following form:

$$\underline{E}(r, \theta, \phi) = \left(\frac{e^{-jk_0 r}}{r} \right) \underline{E}^{\text{FF}}(\theta, \phi)$$

$\underline{E}^{\text{FF}}(\theta, \phi) \equiv$ Normalized far - field electric field vector

In decibels (dB):

$$\text{dB}(\theta, \phi) = 20 \log_{10} \left(\frac{|\underline{E}^{\text{FF}}(\theta, \phi)|}{|\underline{E}^{\text{FF}}(\theta_m, \phi_m)|} \right)$$

$(\theta_m, \phi_m) =$ direction of maximum radiation

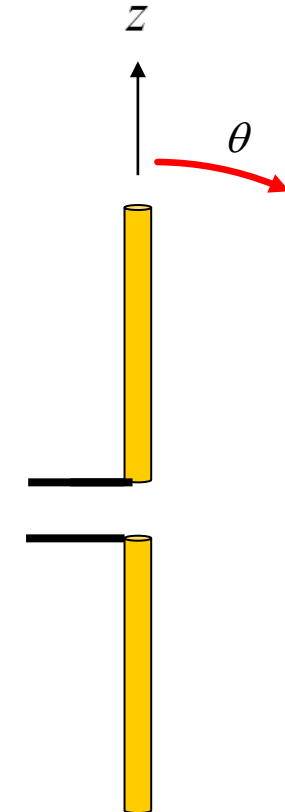
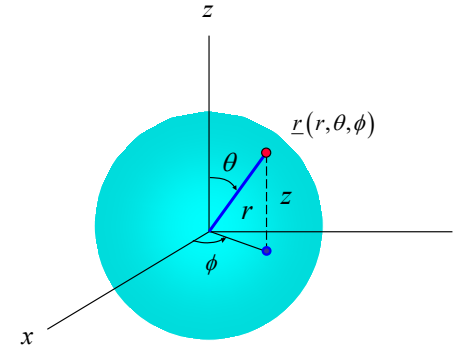
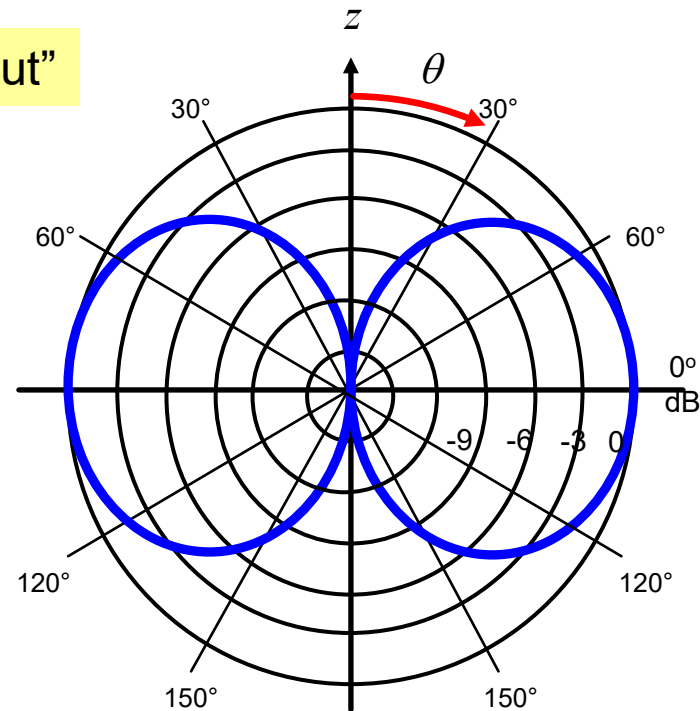
Antenna Radiation (cont.)

A normalized far-field pattern is often shown vs. the angle θ (for a fixed angle $\phi = \phi_0$) in polar coordinates.

$$\text{dB}(\theta, \phi_0) = 20 \log_{10} \left(\frac{|E^{\text{FF}}(\theta, \phi_0)|}{|E^{\text{FF}}(\theta_m, \phi_0)|} \right)$$

The subscript “ m ” denotes the beam maximum.

An “elevation cut”



Antenna Radiation (cont.)

A normalized far-field pattern is often shown vs. the angle ϕ (for a fixed angle $\theta = \theta_0$) in polar coordinates.

$$\text{dB}(\theta, \phi_0) = 20 \log_{10} \left(\frac{|\underline{E}^{\text{FF}}(\theta_0, \phi)|}{|\underline{E}^{\text{FF}}(\theta_0, \phi_m)|} \right)$$

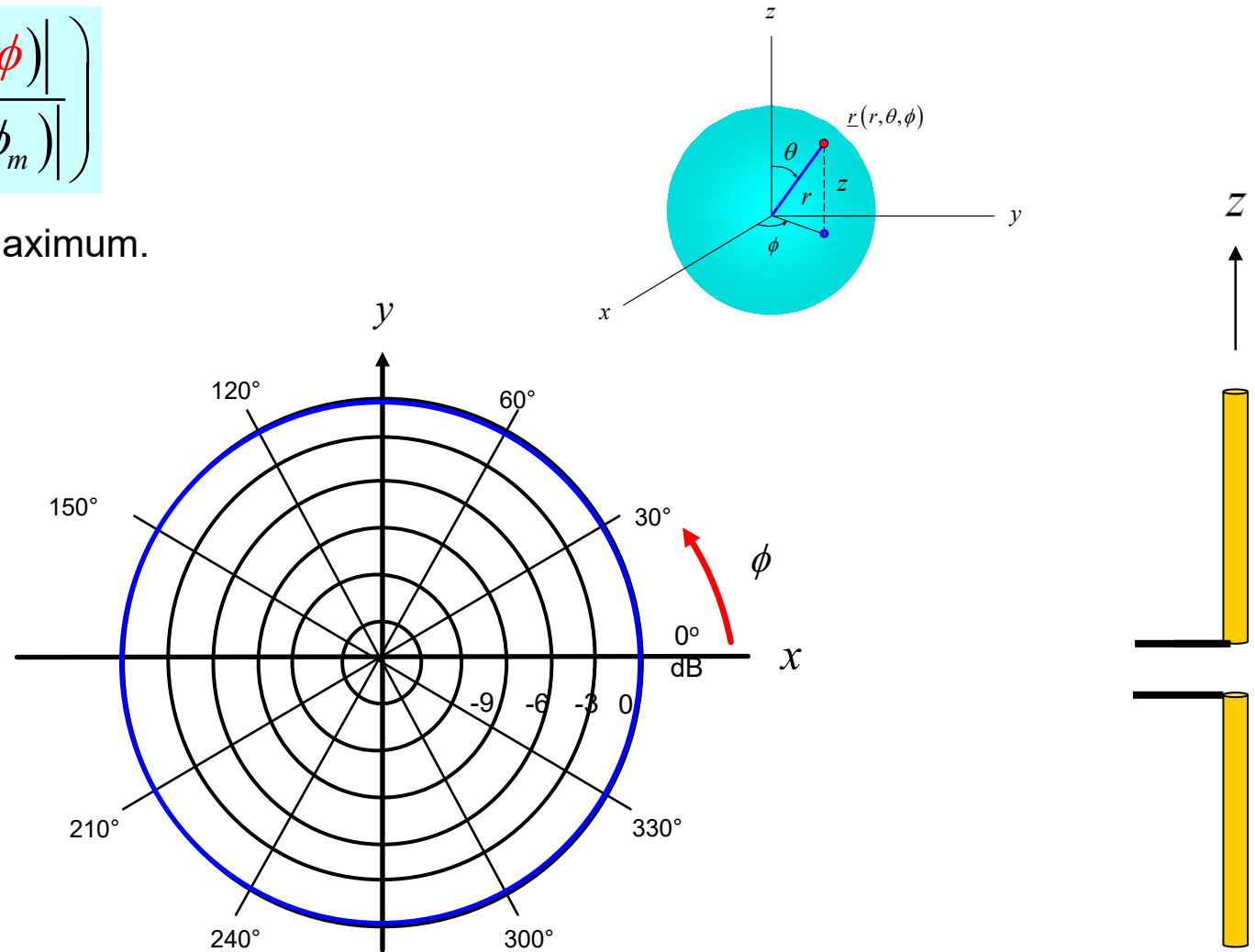
The subscript “ m ” denotes the beam maximum.

An “azimuth cut”

(Usually we use the horizontal plane ($\theta_0 = 90^\circ$.)

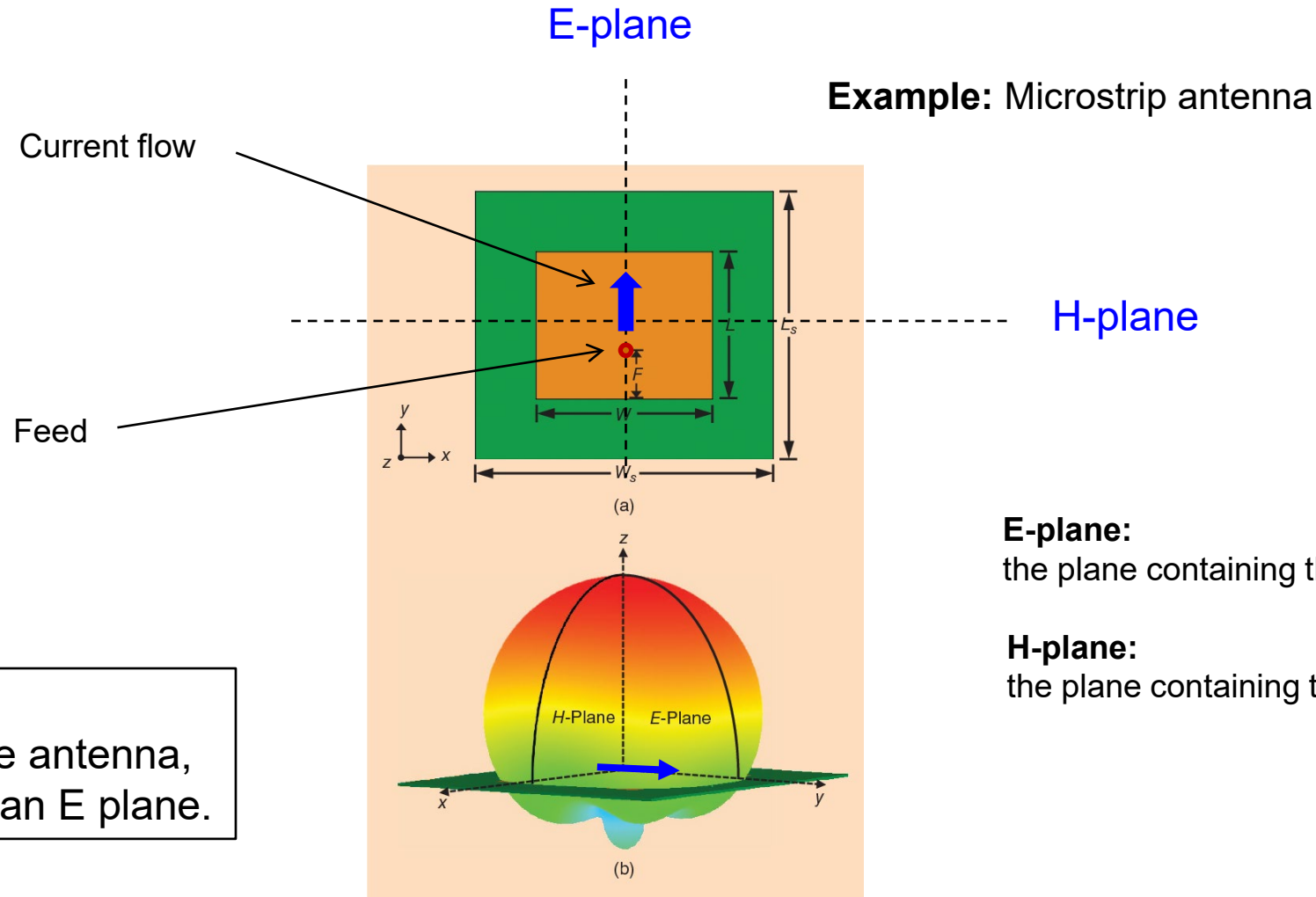
Note:

For the vertical dipole antenna the azimuth pattern is constant.



Antenna Radiation (cont.)

E-plane and H-plane



Note:

For the vertical dipole antenna, every elevation cut is an E plane.

Radiated Power

The Poynting vector in the far field is

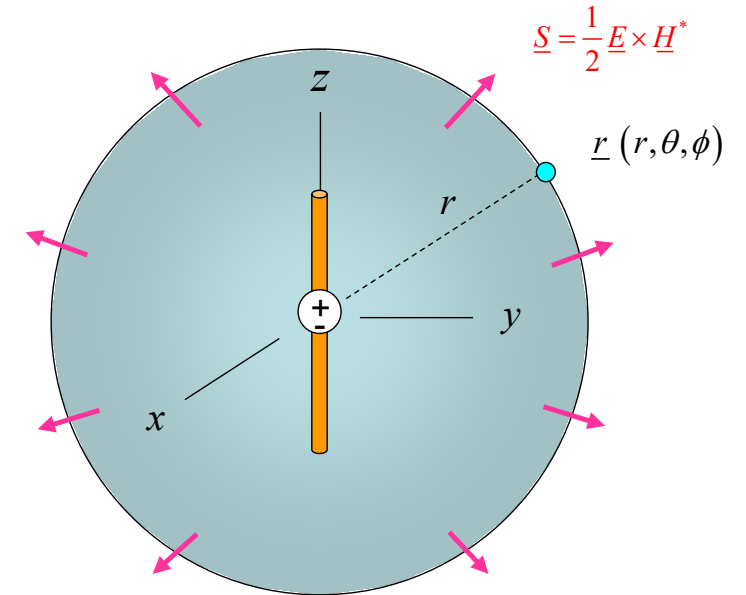
$$\underline{S}(r, \theta, \phi) = \hat{r} \left(\frac{|\underline{E}^{\text{FF}}(\theta, \phi)|^2}{2\eta_0} \right) \left(\frac{1}{r^2} \right) [\text{W/m}^2]$$

The total power radiated (in Watts) is then given by

$$P_{\text{rad}} = \int_0^{2\pi} \int_0^\pi (\underline{S} \cdot \hat{r}) r^2 \sin \theta d\theta d\phi = \int_0^{2\pi} \int_0^\pi \left(\frac{|\underline{E}^{\text{FF}}(\theta, \phi)|^2}{2\eta_0} \right) \sin \theta d\theta d\phi$$

Hence, we have:

$$P_{\text{rad}} = \frac{1}{2\eta_0} \int_0^{2\pi} \int_0^\pi |\underline{E}^{\text{FF}}(\theta, \phi)|^2 \sin \theta d\theta d\phi$$



Directivity

The **directivity** of the antenna in the directions (θ, ϕ) is defined as

$$D(\theta, \phi) \equiv \frac{S_r(\theta, \phi)}{P_{\text{rad}} / (4\pi r^2)} \quad r \rightarrow \infty$$

The directivity (in a particular direction) is the ratio of the actual power density radiated in that direction to the power density that would be radiated in that direction if the antenna were an isotropic radiator (radiates equally in all directions).

In dB,

$$D_{\text{dB}}(\theta, \phi) = 10 \log_{10} D(\theta, \phi)$$

Note: The directivity is sometimes referred to as the “directivity with respect to an isotropic radiator.”

Directivity (cont.)

The directivity is then directly expressed in terms of the far field pattern:

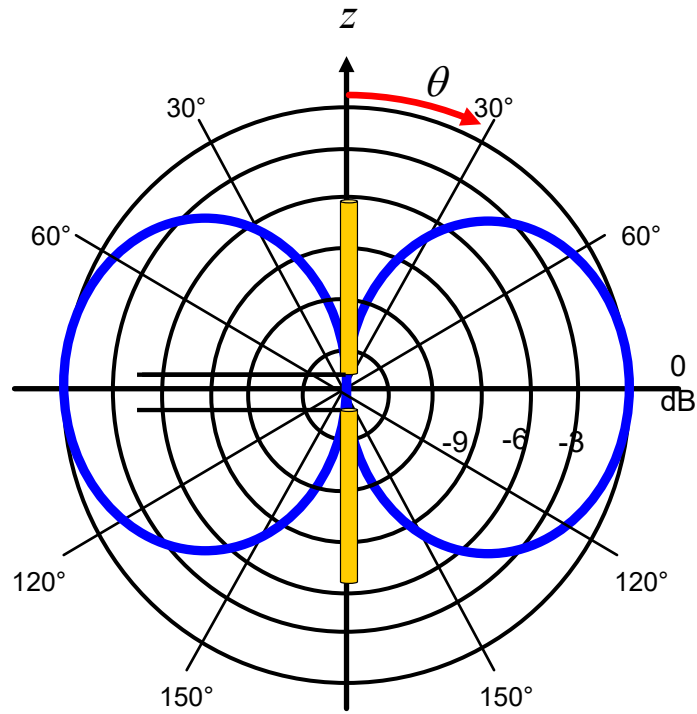
$$D(\theta, \phi) = \frac{4\pi |\underline{E}^{\text{FF}}(\theta, \phi)|^2}{\int_0^{2\pi} \int_0^\pi |\underline{E}^{\text{FF}}(\theta, \phi)|^2 \sin \theta d\theta d\phi}$$

$$\left\{ \begin{array}{l} D(\theta, \phi) \equiv \frac{S_r(\theta, \phi)}{P_{\text{rad}} / (4\pi r^2)} \quad r \rightarrow \infty \\ \underline{S}(r, \theta, \phi) = \hat{r} \left(\frac{|\underline{E}^{\text{FF}}(\theta, \phi)|^2}{2\eta_0} \right) \left(\frac{1}{r^2} \right) [\text{W/m}^2] \\ P_{\text{rad}} = \frac{1}{2\eta_0} \int_0^{2\pi} \int_0^\pi |\underline{E}^{\text{FF}}(\theta, \phi)|^2 \sin \theta d\theta d\phi \end{array} \right.$$

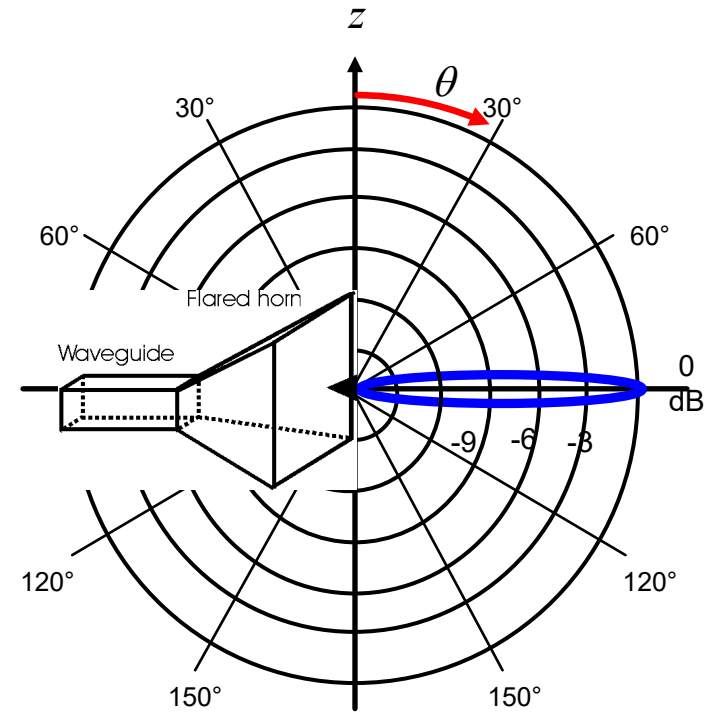
Usually, the directions are chosen to corresponds to the main beam direction:

$$D = D(\theta_m, \phi_m)$$

Directivity (cont.)



Antenna with moderate directivity
(e.g., dipole)



Antenna with high directivity
(e.g., horn or dish)

Directivity (cont.)

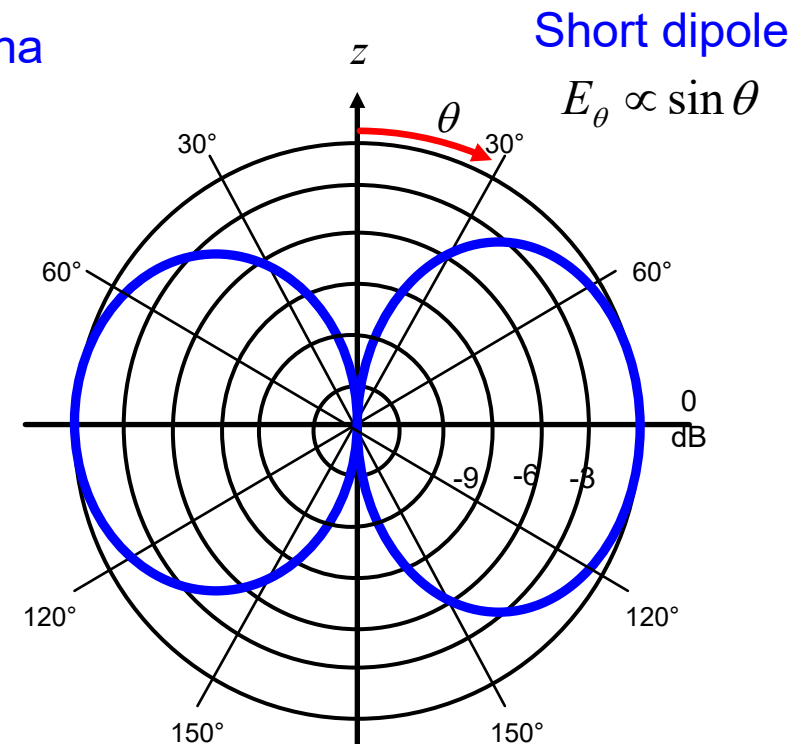
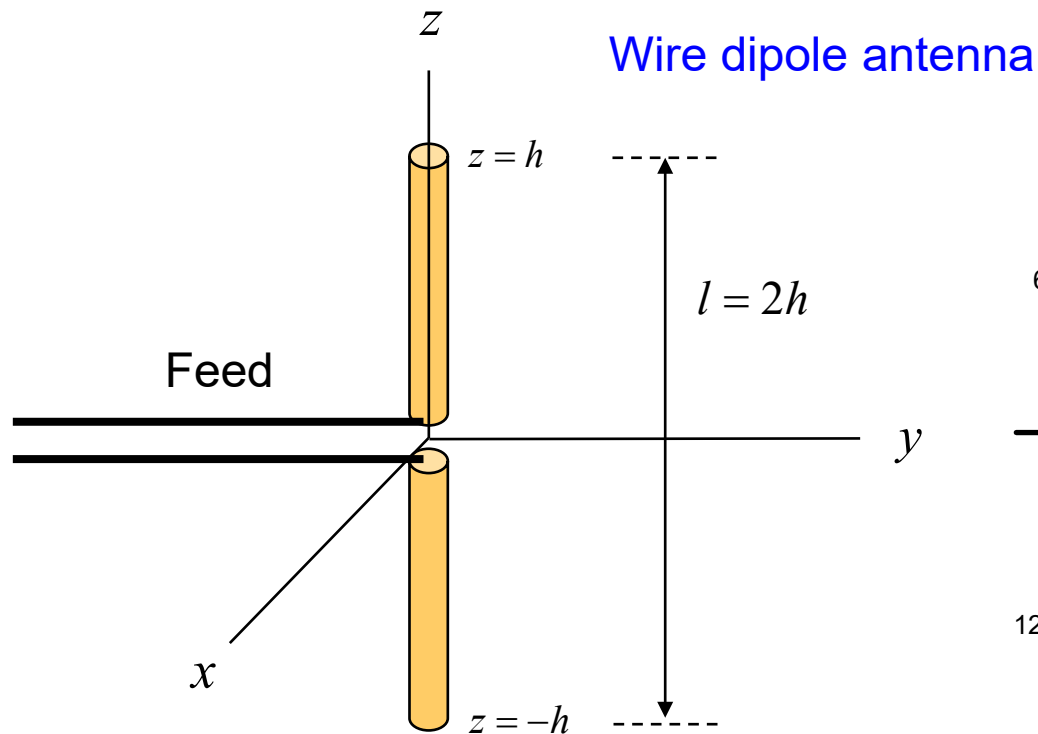
Two Common Cases: Dipole Antennas

$$\theta_m = \pi / 2$$

Short dipole wire antenna ($l \ll \lambda_0$): $D = 1.5$

$$D = D_{\max} = D(\pi / 2, \phi)$$

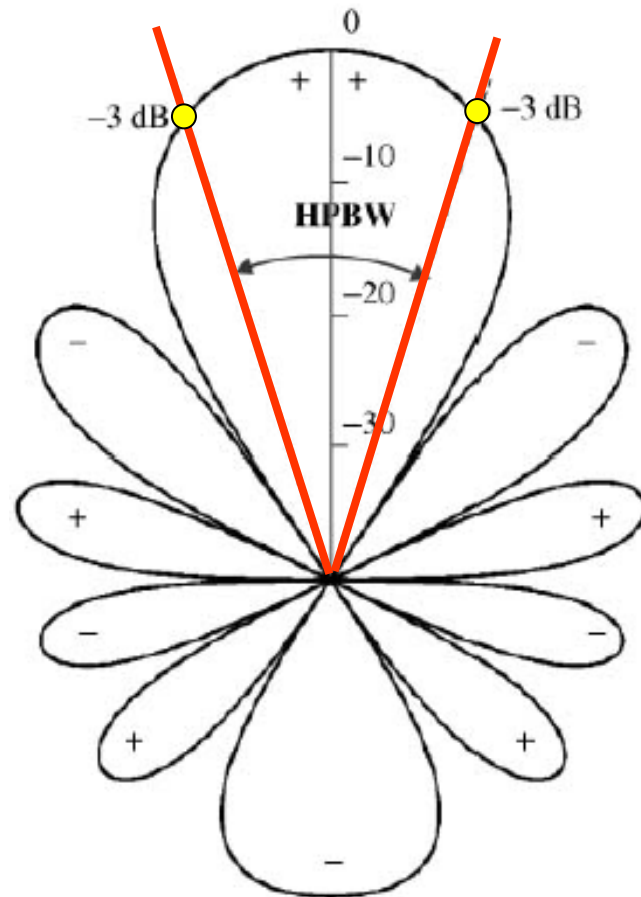
Resonant half-wavelength dipole wire antenna ($l = \lambda_0 / 2$): $D = 1.643$



Beamwidth

The beamwidth measures how narrow the beam is (the narrower the beamwidth, the higher the directivity).

HPBW = half-power beamwidth



At the “half-power” points:

The power density is down by a factor of $1/2$.

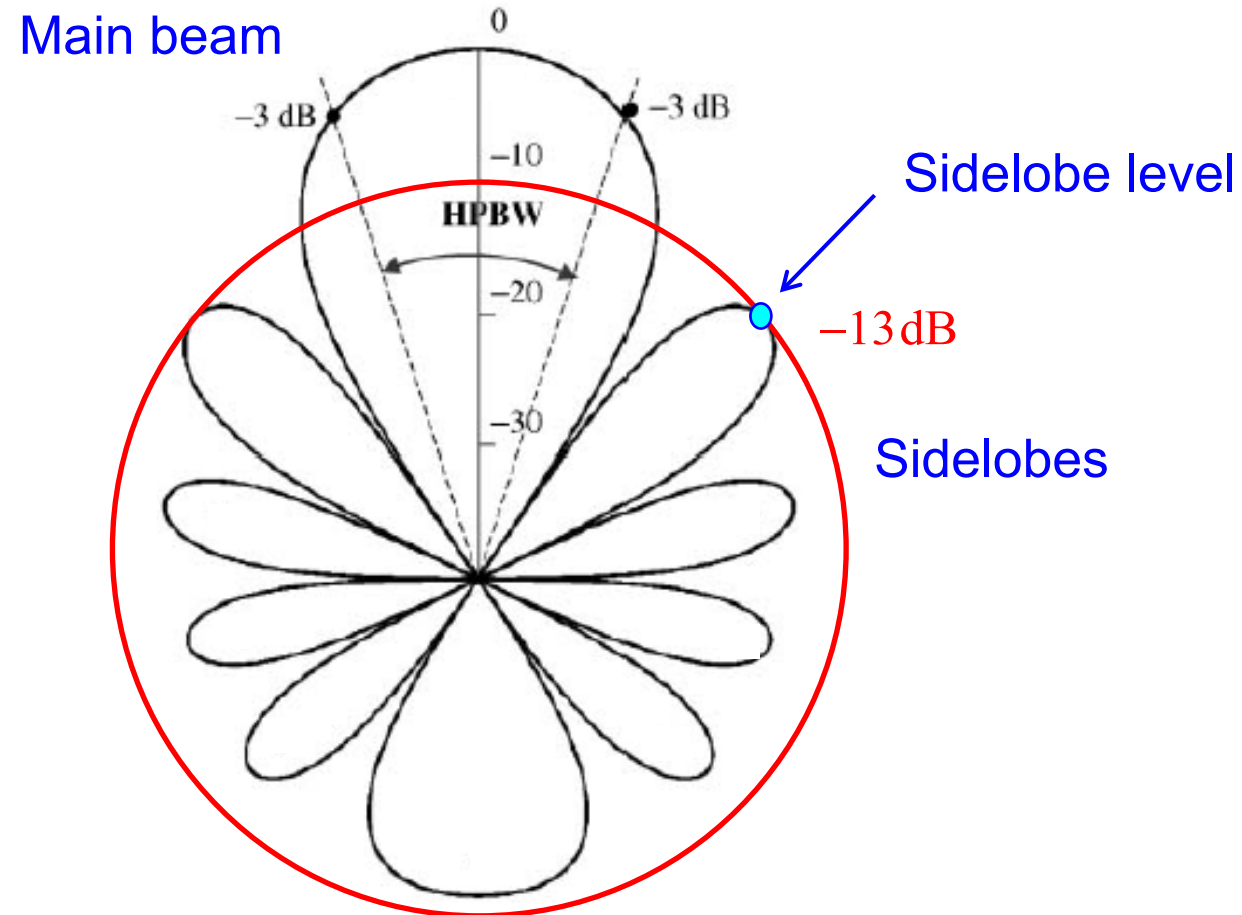
The field is down by a factor of $1/\sqrt{2} = 0.707$.

In dB, we are down by 3 dB.

Sidelobe Level

The sidelobe level measures how large the sidelobes are.

In this example the sidelobe level is about -13 dB.



Gain and Efficiency

The radiation efficiency of an antenna is defined as:

$$e_r \equiv \frac{P_{\text{rad}}}{P_{\text{in}}}$$

P_{rad} = power radiated by the antenna

P_{in} = power input to the antenna

The gain of an antenna in the directions (θ, ϕ) is defined as

$$G(\theta, \phi) \equiv e_r D(\theta, \phi)$$

In dB, we have

$$G_{\text{dB}}(\theta, \phi) = 10 \log_{10} G(\theta, \phi)$$

Gain and Efficiency (cont.)

The gain tells us how strong the radiated power density is in a certain direction, for a given amount of input power.

Recall that

$$D(\theta, \phi) \equiv \frac{S_r(\theta, \phi)}{P_{\text{rad}} / (4\pi r^2)} \quad r \rightarrow \infty$$

Therefore, in the far field:

$$S_r(\theta, \phi) = \left[P_{\text{rad}} / (4\pi r^2) \right] D(\theta, \phi)$$



$$S_r(\theta, \phi) = \left[e_r P_{\text{in}} / (4\pi r^2) \right] D(\theta, \phi)$$



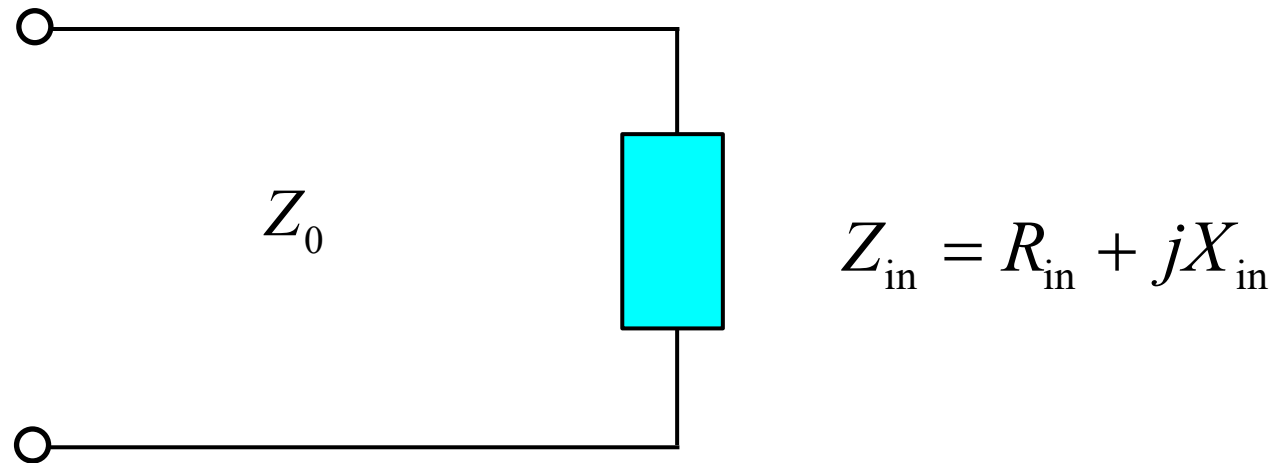
$$S_r(\theta, \phi) = \left[P_{\text{in}} / (4\pi r^2) \right] G(\theta, \phi)$$

Recall: $e_r \equiv \frac{P_{\text{rad}}}{P_{\text{in}}}$

Recall: $G(\theta, \phi) \equiv e_r D(\theta, \phi)$

Input Impedance

The antenna acts like a load impedance during transmit.



At resonance, the input reactance X_{in} is zero (the desired situation).

Note:

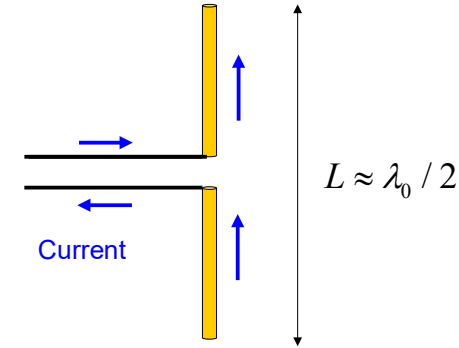
We usually want a match between the input impedance and the characteristic impedance Z_0 of the feeding transmission line, to avoid reflection.

Input Impedance (cont.)

At resonance:

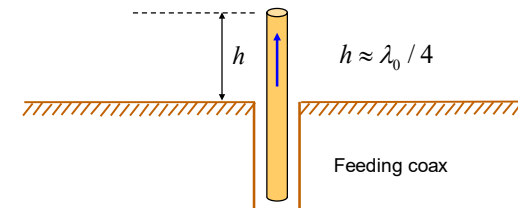
Dipole:

$$Z_{\text{in}} \approx 73 \, [\Omega]$$



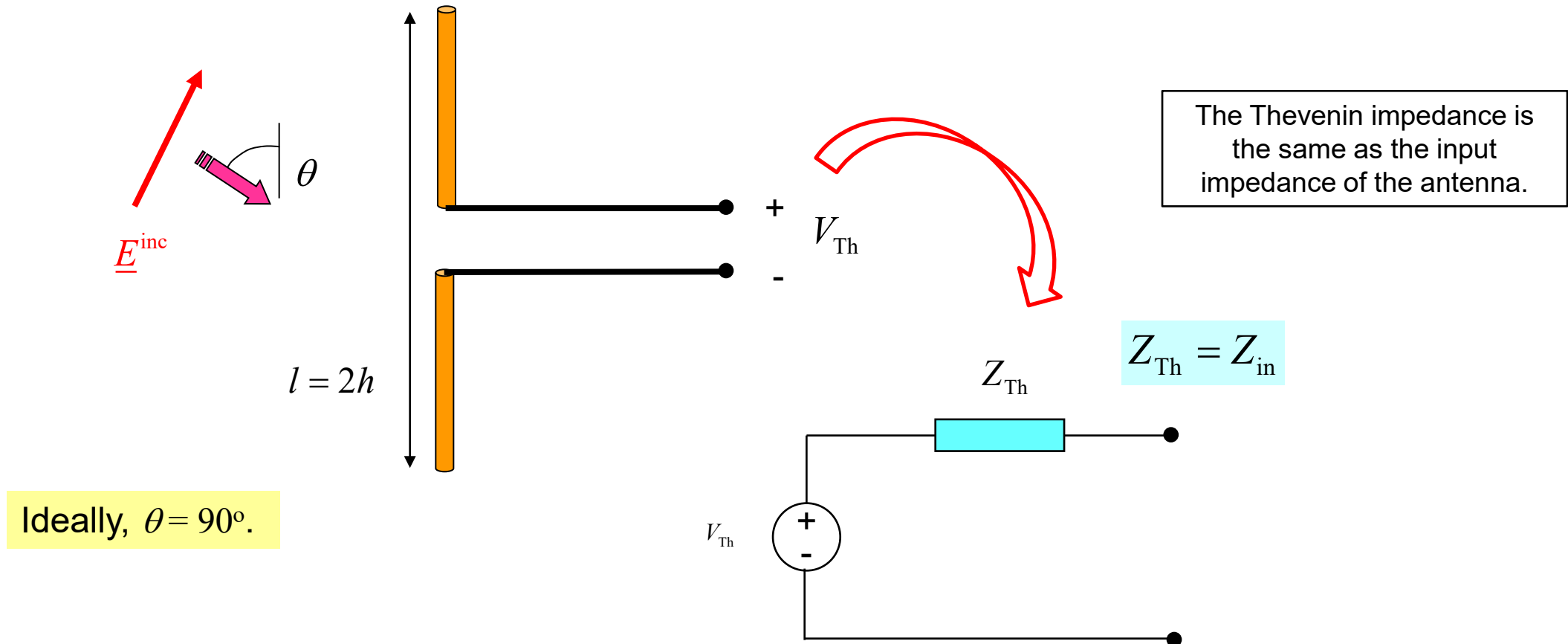
Monopole:

$$Z_{\text{in}} \approx 36.5 \, [\Omega]$$



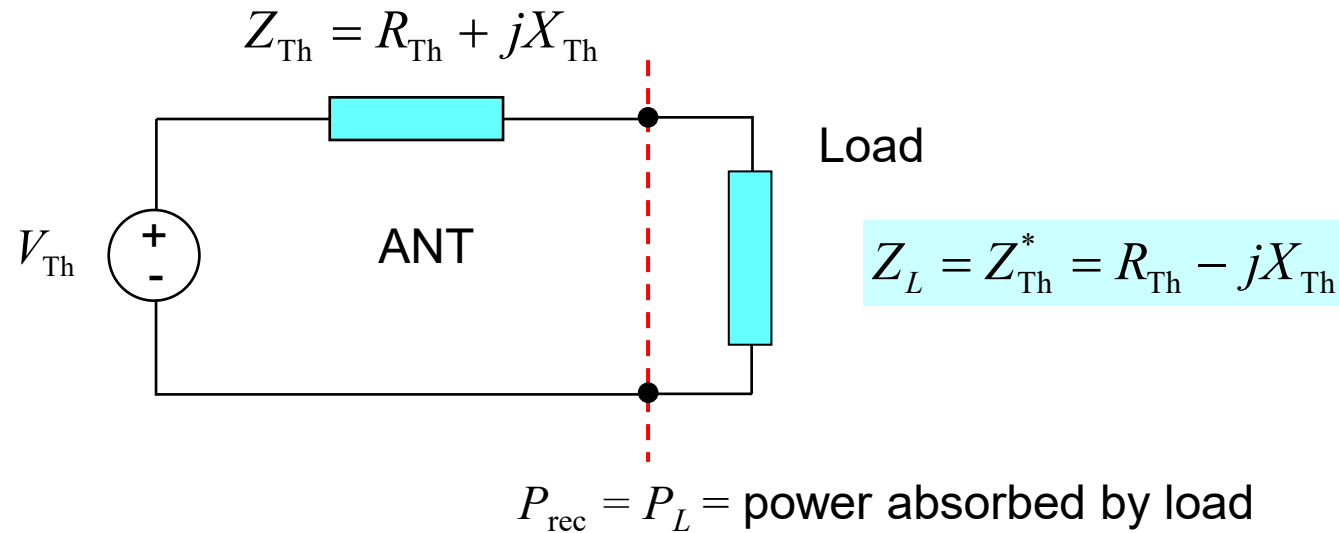
Receive Antenna

The Thévenin equivalent circuit of an antenna being used as a receive antenna is shown below.



Receive Antenna (cont.)

The power received by an optimum conjugate-matched load:



For a resonant dipole wire antenna:

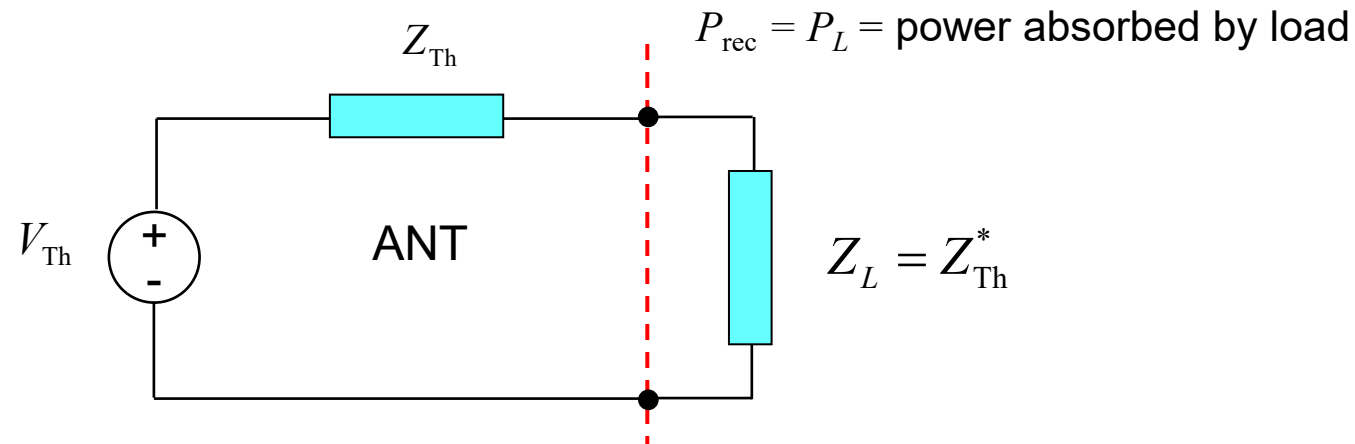
$$X_{Th} = X_{in} = 0$$

$$R_{Th} = R_{in} = 73 [\Omega] \quad \Rightarrow \quad R_{Th} = 73 [\Omega] \quad \Rightarrow \quad R_L^{opt} = 73 [\Omega]$$

Receive Antenna (cont.)

We can find the power received using an **effective area**.

Receive circuit: Assume an optimum conjugate-matched load:



$$P_{rec} = A_{eff} P_d^{inc}$$

A_{eff} = effective area of antenna

P_d^{inc} = power density of incident wave $\left[\text{W/m}^2 \right]$

Receive Antenna (cont.)

We have the following general formula*:

$$A_{\text{eff}} = G \left(\frac{\lambda_0^2}{4\pi} \right)$$

$G = G(\theta, \phi)$ = gain of antenna in direction (θ, ϕ)

(Usually, we assume that (θ, ϕ) is in the main beam direction.)

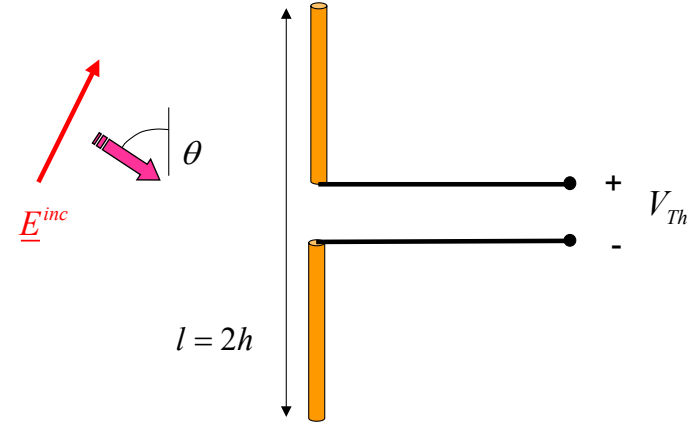
*A derivation is given in the following book: C. A. Balanis, *Antenna Engineering*, 3rd Ed., 2016, Wiley.

Receive Antenna (cont.)

Effective area of a lossless resonant half-wave dipole antenna:

Assuming normal incidence ($\theta = 90^\circ$):

$$\begin{aligned} A_{\text{eff}} &= G \left(\frac{\lambda_0^2}{4\pi} \right) \\ &= 1.643 \left(\frac{\lambda_0^2}{4\pi} \right) \quad (D = D_{\text{max}} = 1.643) \\ &= 1.643 \left(\frac{(2l)^2}{4\pi} \right) \quad (l = \lambda_0 / 2) \end{aligned}$$



Assume lossless antenna:
 $e_r = 100\%$
($G = D$)

Hence:

$$A_{\text{eff}} = 0.523 l^2$$

Note:

The dipole will receive more power at a lower frequency (larger l), assuming the same incident power.

Example with Wire Antennas

Example

Find the received power P_{rec} in the example below, assuming that the receiver is connected to an optimum conjugate-matched load.

$$f = 1 \text{ [GHz]} \quad (\lambda_0 = 0.29979 \text{ [m]})$$

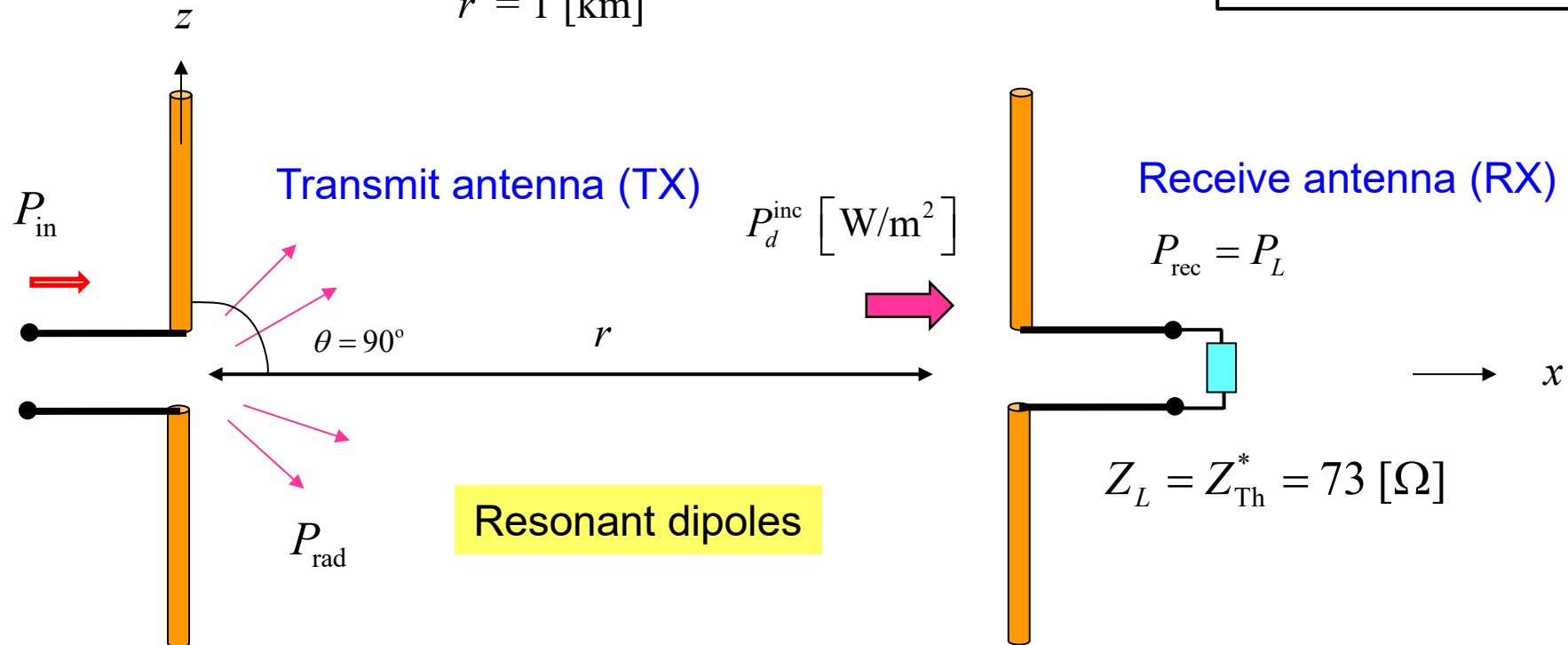
$$P_{\text{in}} = 10 \text{ [W]}$$

$$r = 1 \text{ [km]}$$

Assume lossless antennas:

$$e_r = 100\%$$

$$(G = D)$$



Example with Wire Antennas (cont.)

$$P_{\text{rec}} = A_{\text{eff}} P_d^{\text{inc}}$$

Gain of RX $\downarrow$

$$A_{\text{eff}} = 1.643 \left(\frac{\lambda_0^2}{4\pi} \right)$$

Gain of TX $\downarrow$

$$P_d^{\text{inc}} = \frac{P_{\text{in}}}{4\pi r^2} (1.643)$$


Hence:

$$P_{\text{rec}} = \left[1.643 \left(\frac{\lambda_0^2}{4\pi} \right) \right] \left[\frac{P_{\text{in}}}{4\pi r^2} (1.643) \right]$$

The result is:

$$P_{\text{rec}} = 1.54 \times 10^{-8} \text{ [W]}$$

Receive Antenna (cont.)

Effective area of dish (reflector) antenna

In the maximum gain (main beam) direction:

$$A_{\text{eff}} = A_{\text{phy}} e_{\text{ap}}$$

Now it is the effective area that we know, and from this we can calculate the gain.

A_{phy} = physical area of dish
 e_{ap} = “aperture efficiency”

$$A_{\text{eff}} = G \left(\frac{\lambda_0^2}{4\pi} \right)$$

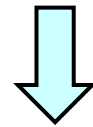


The aperture efficiency is usually less than 1 (less than 100%).

Gain of Dish Antenna

Dish antenna: Obtaining a higher gain means having a larger dish.

$$G = A_{\text{eff}} \left(\frac{4\pi}{\lambda_0^2} \right)$$



$$A_{\text{eff}} = A_{\text{phy}} e_{\text{ap}}$$

$$G = 4\pi \left(\frac{A_{\text{phy}}}{\lambda_0^2} \right) e_{\text{ap}}$$



Example with Dish Antenna

Example

A microstrip antenna on a CubeSat with a gain of 8 (9.03 dB) transmits with an input power of 1 [W] at 10.0 GHz from a distance of 50,000,000 [km] (near Mars).

How much power will be received by the NASA Deep Space Network dish at Goldstone, CA, which has a diameter of 70 [m]? Assume an aperture efficiency of 0.75 (75%).

Express answer in Watts and in dBm (dB relative to a milliwatt).

Note : $P_{\text{rec}}^{\text{dBm}} \equiv 10 \log_{10} \left(\frac{P_{\text{rec}}}{0.001 \text{ [W]}} \right)$

Example with Dish Antenna (cont.)

Example (cont.)

$$P_{\text{rec}} = P_d^{\text{inc}} A_{\text{eff}} = P_d^{\text{inc}} A_{\text{phy}} e_{\text{ap}}$$
$$P_d^{\text{inc}} = \left(\frac{P_{\text{in}}}{4\pi r^2} \right) G_{\text{trans}} \quad \uparrow$$

$$P_{\text{rec}} = 1.014 \times 10^{-19} \text{ [W]}$$

$$P_{\text{rec}}^{\text{dBm}} = -151.3$$

Parameters:

$$r = 5.0 \times 10^{10} \text{ [m]}$$

$$A_{\text{phy}} = \pi (70/2)^2 \text{ [m}^2\text{]}$$

$$e_{\text{ap}} = 0.75$$

$$P_{\text{in}} = 1 \text{ [W]}$$

$$G_{\text{trans}} = 8$$